

Chapter 14 Summary — Evaluation Frameworks and Exploitability Metrics

Research on the possibilities for applying Artificial Intelligence in computer games

Alexander Andreev

September 2026

Contents

14 Chapter 14 — Evaluation Frameworks and Exploitability Metrics	1
14.1 Why evaluating an adaptive agent is hard	2
14.2 The framework: three layers and a fourth question	3
14.3 Layer 2 — rankings disagree, in predictable ways	5
14.4 Layer 3 — confidence, and what AIVAT taught	7
14.5 The joint protocol on Kuhn and Leduc	9
14.6 Approximate best responses and the RRPS score	12
14.7 Three and four players	13
14.8 Where existing evaluation breaks — a checklist	14
14.9 Honest notes, limitations, and where this hands off	15
14.10 Key takeaways for the thesis synthesis	16

14 Chapter 14 — Evaluation Frameworks and Exploitability Metrics

This is a ground-up chapter on how to evaluate game-playing agents that **adapt**: agents that watch their opponents and change their play. It builds the standard toolkit — exact exploitability, population rankings and variance reduction — validates every piece against an independent reference, and then uses it to measure what that toolkit misses. **All experimental numbers were measured** on reproducible runs on three small poker games (two-player Kuhn and Leduc, three-player Kuhn) and on Chapter 11’s four-player So Long Sucker engine, and wherever the game allows they are *exact* rather than sampled. Where a run contradicted what I expected, I keep the expectation and reconcile it with what happened.

Where this sits in the thesis. This chapter is Contribution 3 (evaluation methodology). The September 2026 literature check narrowed that contribution: combining exploitability, population ranking and confidence is not new on its own — a benchmark for repeated rock–paper–scissors already scores return and exploitability together, for one game¹. What no protocol found reports together is (a) the **gain** against a sub-optimal population, (b) the **speed** of adaptation and the **recovery** after an opponent

¹Lancot, M., Schultz, J., Burch, N., Smith, M. O., Hennes, D., Anthony, T. & Pérolat, J. (2023). “Population-based Evaluation in Repeated Rock-Paper-Scissors as a Benchmark for Multiagent Reinforcement Learning.” *Transactions on Machine Learning Research*; arXiv:2303.03196 — 43 bots, 1,000-throw matches; population return, within-population exploitability and their difference as an aggregate score.

switches, and (c) the **worst case**, or with more than two players a coalition-aware substitute — each with confidence bounds, applied unchanged across several imperfect-information games. The chapter builds such a protocol and shows, on concrete agent zoos, where the existing measures break. It also serves the other two contributions: it is the yardstick for Chapter 7’s adaptive agents (Contribution 1) and for the question Chapter 8 left open — what “safe” can mean with three players (Contribution 2).

14.1 Why evaluating an adaptive agent is hard

There are two ways to grade a game-playing agent. The first asks *how badly can it be beaten?* In a two-player zero-sum game the answer is **exploitability**: how far a best-responding opponent can push the agent below the game value. The second asks *how well does it do against the opponents it will meet?* The answers are head-to-head win rates and ratings computed from them, from Elo² to Nash averaging³, α -Rank⁴ and voting-based evaluation⁵. The two families are known to disagree. Two poker bots that were indistinguishable head-to-head were about 1,300 mbb/game apart in local-best-response exploitability⁶, and the Annual Computer Poker Competition used two winner rules for exactly this reason⁷.

An adaptive agent makes the disagreement structural. It leaves equilibrium on purpose, to exploit, so exploitability can only count against it however much it gains. Its payoff against another agent depends on how long they play and on what it has seen, so it has no single entry in the payoff matrix that every population ranking assumes. With three or more players the worst case itself changes meaning: NashConv, the usual N-player generalisation of exploitability, measures distance from *an* equilibrium⁸. In three-player Kuhn poker a player can pass utility to a second player at the third’s expense while everyone plays equilibrium strategies⁹. The only worst-case notion the literature check found for coalitions is the team-maxmin value of adversarial team games¹⁰ — a solution concept, not an evaluation protocol.

A picture helps. Hiring a negotiator, one reference says “the worst deal a ruthless counterpart could force on this person”, another “how they did in their last twenty negotiations”. Neither says how fast they read

²Elo, A. E. (1978). *The Rating of Chessplayers, Past and Present*. New York: Arco.

³Balduzzi, D., Tuyls, K., Pérolat, J. & Graepel, T. (2018). “Re-evaluating Evaluation.” *NeurIPS*; arXiv:1806.02643 — Elo is inflated by copies of beaten agents; Nash averaging is invariant to redundant agents.

⁴Omidshafiei, S., Papadimitriou, C., Piliouras, G., Tuyls, K., Rowland, M., Lespiau, J.-B., Czarnecki, W. M., Lanctot, M., Pérolat, J. & Munos, R. (2019). “ α -Rank: Multi-Agent Evaluation by Evolution.” *Scientific Reports* 9, 9937. DOI 10.1038/s41598-019-45619-9.

⁵Lanctot, M., Larson, K., Bachrach, Y., Marris, L., Li, Z., Bhoopchand, A., Anthony, T., Tanner, B. & Koop, A. (2023). “Evaluating Agents using Social Choice Theory.” *arXiv:2312.03121* (v4, 2025) — Voting-as-Evaluation; maximal lotteries recommended.

⁶Moravčík, M. et al. (2017). “DeepStack: Expert-level artificial intelligence in heads-up no-limit poker.” *Science* 356(6337), 508–513; arXiv:1701.01724 — also reports an 85 % reduction in standard deviation with AIVAT.

⁷Bard, N., Hawkin, J., Rubin, J. & Zinkevich, M. (2013). “The Annual Computer Poker Competition.” *AI Magazine* 34(2), 112–114 — total-bankroll and instant-runoff winner rules; duplicate play.

⁸Lanctot, M. et al. (2019). “OpenSpiel: A Framework for Reinforcement Learning in Games.” *arXiv:1908.09453* — NashConv as the sum of each player’s incentive to deviate; exploitability = NashConv / n.

⁹Szafron, D., Gibson, R. & Sturtevant, N. (2013). “A Parameterized Family of Equilibrium Profiles for Three-Player Kuhn Poker.” *AAMAS*, 247–254.

¹⁰Celli, A. & Gatti, N. (2018). “Computational Results for Extensive-Form Adversarial Team Games.” *AAAI* 32(1). DOI 10.1609/aaai.v32i1.11462; see also Zhang, Y., An, B. & Černý, J. (2021). “Computing Ex Ante Coordinated Team-Maxmin Equilibria in Zero-Sum Multiplayer Extensive-Form Games.” *AAAI* 35(6), 5813–5821.

a new counterpart, whether they notice when the counterpart changes tactics, or whether they can be played by someone who acts naive and then turns. With three parties at the table, neither says what happens when the other two coordinate. The chapter’s protocol asks all of these, with error bars.

14.2 The framework: three layers and a fourth question

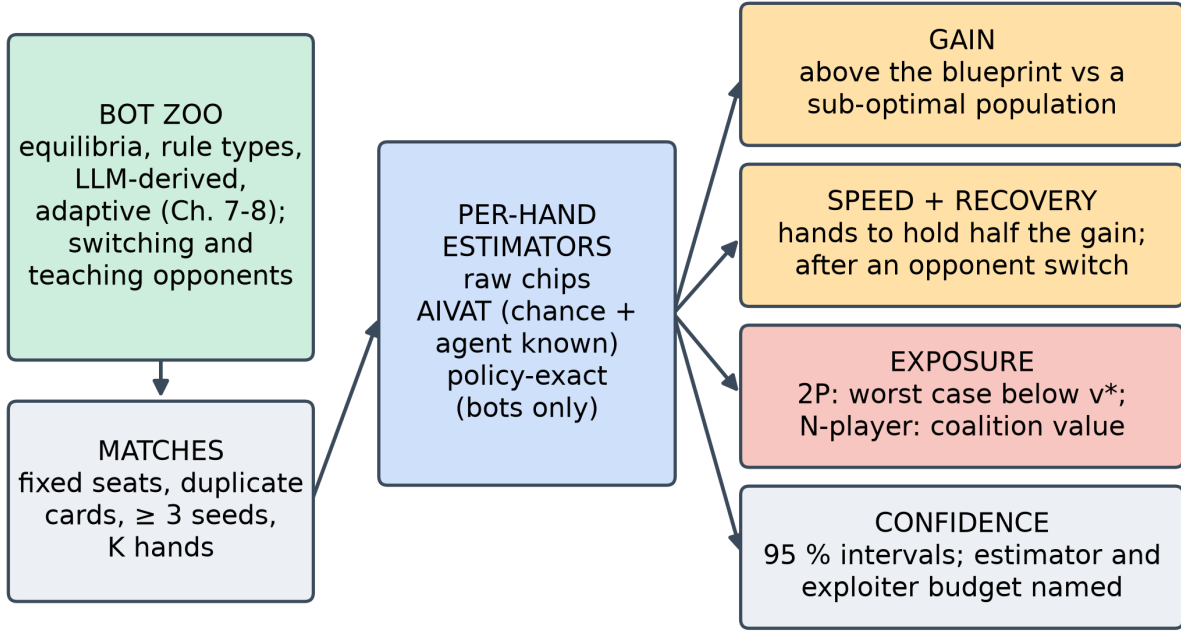
The plan specifies three layers. **Layer 1** is the worst case: exact exploitability in small games, a learned best response in large ones¹¹. **Layer 2** is the population: round-robin payoff matrices, Elo, Nash averaging, the meta-Nash of empirical game analysis (Chapters 9–10), α -Rank, VasE’s maximal lotteries and the transitive/cyclic “spinning top” decomposition¹²¹³. **Layer 3** is confidence: seeds, duplicate dealing and AIVAT¹⁴. The fourth question is the chapter’s own: how an adaptive agent behaves *over time*.

¹¹Timbers, F., Bard, N., Lockhart, E., Lanctot, M., Schmid, M., Burch, N., Schrittwieser, J., Hubert, T. & Bowling, M. (2022). “Approximate Exploitability: Learning a Best Response.” *IJCAI*; arXiv:2004.09677.

¹²Balduzzi, D. et al. (2019). “Open-ended Learning in Symmetric Zero-sum Games.” *ICML*, PMLR 97, 434–443 — the transitive/cyclic decomposition.

¹³Czarnecki, W. M., Gidel, G., Tracey, B., Tuyls, K., Omidshafiei, S., Balduzzi, D. & Jaderberg, M. (2020). “Real World Games Look Like Spinning Tops.” *NeurIPS*; arXiv:2004.09468.

¹⁴Burch, N., Schmid, M., Moravčík, M., Morrill, D. & Bowling, M. (2018). “AIVAT: A New Variance Reduction Technique for Agent Evaluation in Imperfect Information Games.” *AAAI* 32(1); arXiv:1612.06915 — unbiased for any value function (Theorem 1); on Leduc, 48–75 % lower standard deviation for dissimilar strategies.



Context, not a verdict: Elo, Nash averaging, α -Rank (α swept), VasE, spinning top

Figure 1: The joint protocol. A bot zoo — equilibria, rule-based types, LLM-derived strategies, adaptive agents from Chapters 7 and 8, plus switching and teaching opponents — plays fixed-seat duplicate matches over several seeds. Each hand is scored by three estimators, and four readouts are reported together. Population rankings are reported alongside as context.

Everything rests on one exact engine. Each game is enumerated once from OpenSpiel’s own definition into arrays of information sets, sequences and terminal histories. An agent’s strategy at any moment is then a vector, and its expected payoff, best response and exposure are matrix operations. Leduc has 9,457 nodes and 468 information sets per player; building its tree takes 0.04 s. An adaptive agent is treated as a *sequence* of strategies, one per refit, each of which can be scored exactly. The engine reproduces OpenSpiel’s NashConv to 10 decimals on twelve profiles, Chapter 3’s Leduc exploitability (0.811708 after 10 CFR iterations, 0.090002 after 100) and Chapter 7’s best responses to $1e-14$ (`validation.json`).

The four readouts are defined as follows (chips per hand throughout).

- **Gain:** the agent’s expected payoff minus the equilibrium blueprint’s, against the same opponent in the same seat; its **capture** divides this by what the exact best response would gain.
- **Speed and recovery:** *h50* is the first hand at which the capture reaches one half and stays there on average over the next 100 hands; *r50* is the same count after the opponent switches. The adaptive agents refit every 50 hands, so both are resolved to 50 hands.
- **Exposure:** for two players, how far below the game value v^* a best responder could push the policy in force — exploitability applied to the current policy rather than to a fixed agent; plus the realised

loss under a white-box teaching attack. For three players, the **coalition value**: the least the other two can hold the agent to with coordinated strategies.

- **Confidence**: 95 % intervals over seeds, with the estimator named. There are three: raw chips; AIVAT with the agent’s own strategy known; and the *policy-exact* expected payoff of the two policies in force, available only when both are bots.

The zoos have 16 agents on Kuhn and 12 on Leduc. The **equilibrium** blueprint comes from OpenSpiel’s CFR+ (exploitability 9.6e-6 on Kuhn and 8.5e-5 on Leduc). The **rule-based types** are Chapter 7’s (AlwaysPass, AlwaysBet, TightPassive, LooseAggr, Threshold on Kuhn; CallingStation, Maniac, Rock, LoosePassive on Leduc), plus Random. On Kuhn there are also three **strategies extracted from real language models** in Chapter 12 (Qwen2.5-7B-Instruct, gpt-oss-20b and OpenThinker3-7B, served locally by LM Studio 0.4.20, plain prompt, temperature 0.7, read from the action-token log-probabilities; the quantisation of the weights was not recorded). OpenThinker3-7B is a weak stand-in: 34 % of its probability mass at the action token could not be mapped to an action. A **specialist** is a static best response to one weak type. The five **adaptive agents** combine Chapter 7’s models with Chapter 8’s responses:

- *TypeBR*: a Bayesian posterior over the types, best-responded to;
- *DirBR*: the continuous Dirichlet model, best-responded to;
- *DirBR-CP*: DirBR with Chapter 7’s change-point reset;
- *RNR(0.5)*: the continuous model inside a restricted Nash response with $p = 0.5$ ¹⁵;
- *BestEq*: the best equilibrium against the model — the most it can earn while never falling below v^* , Ganzfried and Sandholm’s best-equilibrium baseline¹⁶.

Chapter 8 solved the last two with a constraint-generation loop that did not converge on full Leduc within its caps. Here they are solved with the sequence-form dual linear program^{17,18}, in which the opponent’s best response enters as a dual variable. One LP per method solves full Leduc in 0.02–0.04 s and reproduces Chapter 8’s Kuhn results. The only difference is a 1.9e-3 gap in the best equilibrium, which comes from the 5e-4 slack Chapter 8’s solver applies to its floor.

14.3 Layer 2 — rankings disagree, in predictable ways

Each pair of agents plays 2,000 hands per match, in both seats with duplicate cards, over 10 seeds; pairs of stationary agents are evaluated exactly. Elo is fitted to the probability of finishing a 100-hand session ahead. The round-robin gives a clear and uncomfortable picture (`population_{kuhn,leduc}.json`).

¹⁵Johanson, M., Zinkevich, M. & Bowling, M. (2007). “Computing Robust Counter-Strategies.” *NIPS 2007* — Restricted Nash Response.

¹⁶Ganzfried, S. & Sandholm, T. (2015). “Safe Opponent Exploitation.” *ACM Transactions on Economics and Computation* 3(2), Article 8. DOI 10.1145/2716322 — the best-equilibrium strategy is their baseline; their safe algorithms risk only gifts already won.

¹⁷Koller, D., Megiddo, N. & von Stengel, B. (1996). “Efficient Computation of Equilibria for Extensive Two-Person Games.” *Games and Economic Behavior* 14(2), 247–259.

¹⁸von Stengel, B. (1996). “Efficient Computation of Behavior Strategies.” *Games and Economic Behavior* 14(2), 220–246 — the sequence form.

DirBR	10	2	1	2	6	4	4	5
RNR(0.5)	3	1	2	1	5	3	1	4
TypeBR	8	6	3	7	3	5	6	3
BR-Rock	5	5	4	6	4	5	5	7
DirBR-CP	12	7	5	5	8	5	7	6
BestEq	1	3	6	3	1	1	2	2
Nash	1	4	7	4	1	2	3	1
Rock	4	8	8	8	7	8	8	7
LoosePassive	7	9	9	9	10	9	9	7
CallingStation	6	11	10	10	11	11	9	7
Maniac	9	10	11	11	9	10	9	7
Random	11	12	12	12	12	12	9	7

Exploit. Elo Pop. return RRPS score Nash avg. VasE α -Rank 0.1 α -Rank 100
 ranking method (rank 1 = best)

Figure 2: Rank of every Leduc agent under eight methods (1 = best), agents ordered by population return. RNR(0.5) and DirBR lead Elo, population return and the RRPS score; BestEq and Nash share first place on exploitability and Nash averaging, BestEq leads VasE, and Nash leads α -Rank at $\alpha = 100$.

The rankings fall into two camps. **Exploitability, Nash averaging and VasE** favour agents that never lose. On Leduc the maximum-entropy Nash mixture of the meta-game is BestEq 0.88, Nash 0.11 and TypeBR 0.01. RNR(0.5) and DirBR earn the highest population return (+0.807 and +0.890 chips/hand) yet get zero weight, because each loses a little to the equilibria. **Elo, population return and the RRPS score** favour agents that beat many opponents: Elo’s order on Leduc is RNR(0.5) 1892, DirBR 1848, BestEq 1779, Nash 1763. The Kendall correlation between the Elo and exploitability rankings is 0.45 on Kuhn and 0.36 on Leduc. Neither camp is wrong about its own question. The point is that for an exploiting agent the two questions have opposite answers, and a single ranking silently picks one.

Three more predicted failure modes appear. **Clones inflate Elo.** Adding up to eight copies of the weakest bot widens DirBR’s Elo lead over Nash from +166 to +181 points on Kuhn and from +84 to +113 on Leduc. No Nash-averaged skill moves by more than $6e-6$, which is the invariance Balduzzi et al. designed Nash averaging to have³. **α -Rank depends on α .** As α grows from 0.01 to 100, three different agents lead on Kuhn (DirBR, RNR(0.5), Nash) and four on Leduc (DirBR, RNR(0.5), BestEq, Nash). **The horizon matters.** Truncating the same runs to 100 hands puts TypeBR first in Elo on Kuhn and the *static* specialist BR-Rock first on Leduc; from 300 hands on, DirBR-CP, then DirBR (Kuhn)

and RNR(0.5) (Leduc) lead. An adaptive agent’s place in a meta-game is only defined once the match length is fixed.

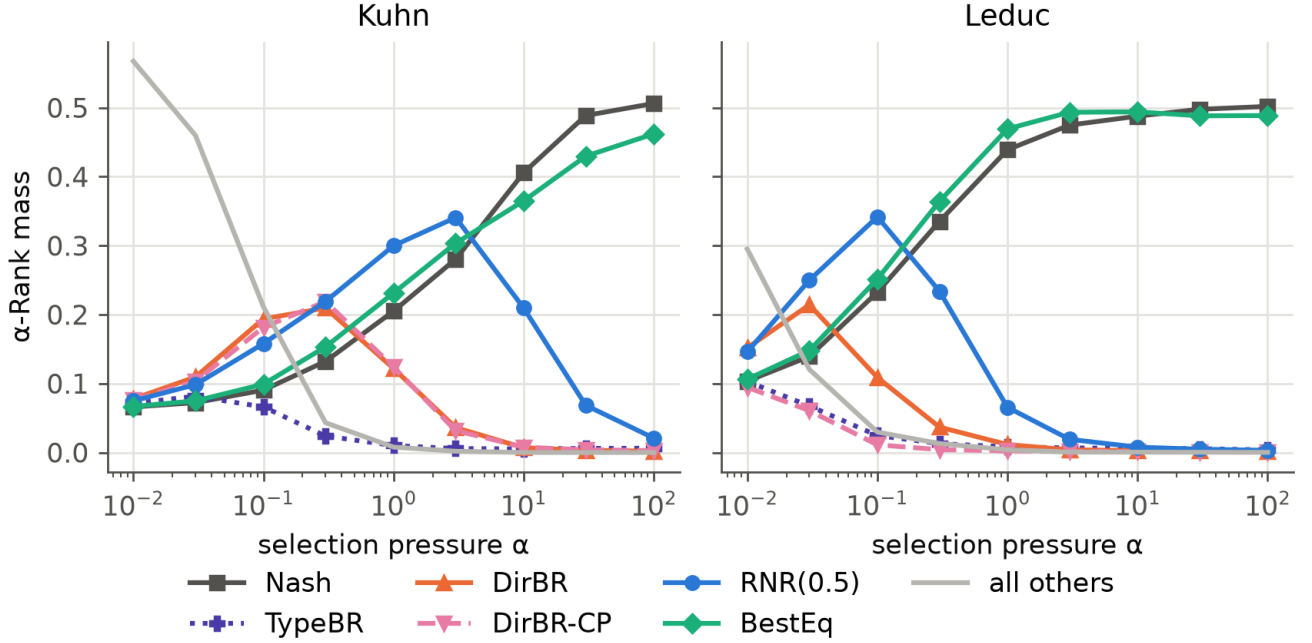


Figure 3: α -Rank mass of the main agents as the selection pressure α grows. On Kuhn the top agent is DirBR for $\alpha \leq 0.1$, RNR(0.5) for $0.3 \leq \alpha \leq 3$ and Nash for $\alpha \geq 10$; on Leduc it moves from DirBR to RNR(0.5), to BestEq and finally to Nash.

Ranking confidence is uneven too. Resampling the 10 seeds 200 times, Elo keeps its Kuhn winner in every resample and α -Rank at $\alpha = 10$ in 22 %. Near-tied equilibrium agents make α -Rank’s top unstable at this sample size — the effect Rowland et al. analyse for noisy payoffs¹⁹. The LLM-derived strategies behave like any other fixed bot, and an arena-style rating misorders them relative to their worst case. Elo puts the Qwen2.5-7B strategy 7th of 16 and the rule-based Threshold bot 10th, although Qwen’s exploitability is 0.179 chips/hand against Threshold’s 0.118 (Chapter 12 reported 0.357 for Qwen as NashConv; the convention here is NashConv/2). LLM arenas that rank by rating alone inherit this.

14.4 Layer 3 — confidence, and what AIVAT taught

AIVAT corrects each hand’s result with control variates for chance events and for the actions of any player whose strategy is known, plus “imaginary observations” over that player’s other possible private cards¹⁴. The paper proves the estimate unbiased for any value function. Because every term depends only on the observed terminal history and the known strategies, this chapter tabulates the AIVAT value of every terminal once per strategy. That gives the estimator’s *exact* mean and variance, with no sampling.

¹⁹Rowland, M., Omidshafiei, S., Tuyls, K., Pérolat, J., Valko, M., Piliouras, G. & Munos, R. (2019). “Multiagent Evaluation under Incomplete Information.” *NeurIPS*; arXiv:1909.09849.

The first implementation had a real bug, and the check that caught it is worth keeping. With both strategies known and a perfect value function, AIVAT’s variance must be zero. It was not. The known player’s own card deal had received a chance-correction term, although the imaginary observations already average over that card; its luck was being counted twice. The paper’s worked example has no term at that deal. Removing it restored exactly zero variance and made the realistic version, with only the agent’s own strategy known, better than chance-only correction everywhere.

With blueprint self-play values as the value function, AIVAT is unbiased to $1e-15$ on every Nash-versus-zoo pair. The plan’s targets — variance divided by at least 5 on Kuhn and 10 on Leduc — are met for 18 of 20 Kuhn pairs (median factor 48) and 7 of 12 Leduc pairs (median 12.2, minimum 6.9). The misses are opponents whose play the self-play value function predicts badly. This is consistent with the paper’s own Leduc table, where dissimilar strategies saw 48–75 % reductions in standard deviation. Two caveats from recent work carry over: AIVAT cannot be used for humans, whose strategy is unknown²⁰, and its value function must be fixed before the data are seen²¹ — here it is the blueprint, computed before any match.

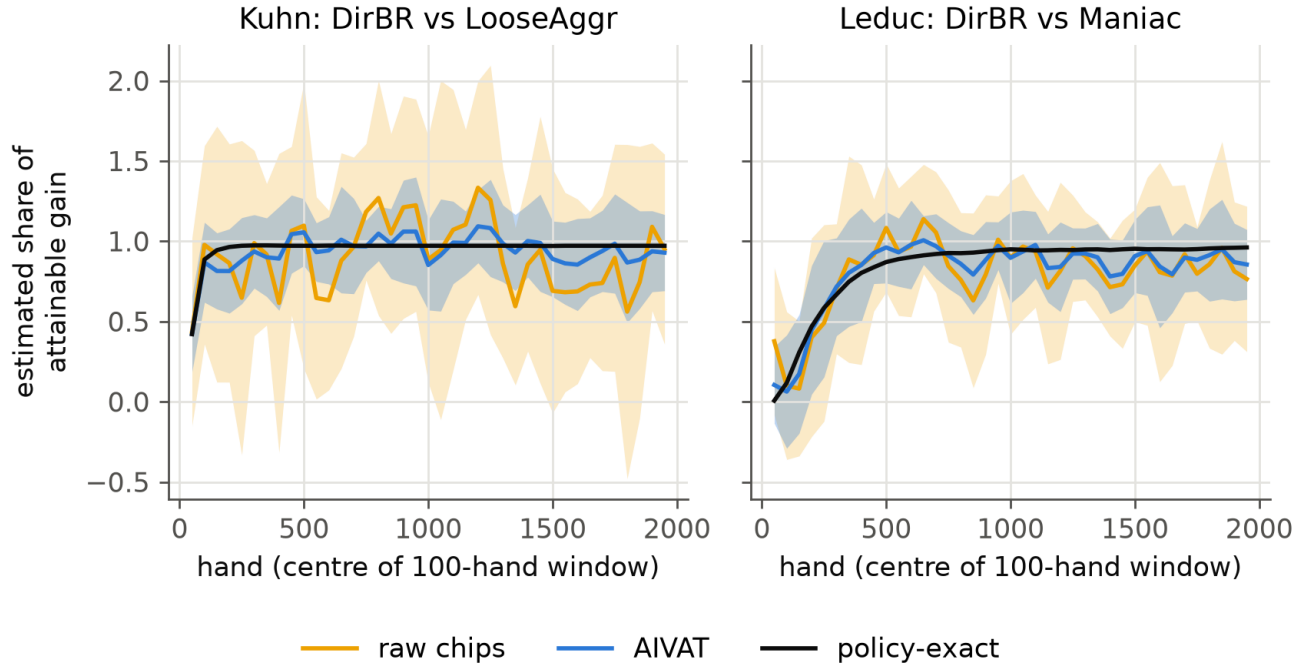


Figure 4: Share of the attainable gain estimated in 100-hand windows from raw chips, from AIVAT and from the exact policy values, for DirBR against LooseAggr (Kuhn) and Maniac (Leduc). Lines are means and bands ± 1 SD over 20 matches.

Measuring *adaptation* rather than a final win rate raises the price sharply. The quantity of interest lives in short windows along the match. To estimate the capture of one 100-hand window to ± 0.25 with 95

²⁰Brown, N. & Sandholm, T. (2019). “Superhuman AI for multiplayer poker.” *Science* 365(6456), 885–890. DOI 10.1126/science.aay2400 — 48 mbb/game, standard error 25, over 10,000 hands; AIVAT cannot be applied to the human players.

²¹Kim, J. & Sandholm, T. (2026). “Heuristic Pathologies and Further Variance Reduction via Uncertainty Propagation in the AIVAT Family of Techniques.” *arXiv:2605.14261*.

% confidence, the median agent–opponent pair needs 3,383 hands with raw chips on Kuhn and 379 with AIVAT; on Leduc, 1,313 and 354. The same holds on real data. Chapter 13 parsed the 10,000 released hands of Pluribus against five professionals. From the per-hand results alone, Pluribus’s win rate is -70.9 ± 172.8 mbb/hand (95 % interval), a standard error of 88 mbb. With AIVAT and Pluribus’s strategy, Brown and Sandholm report 48 mbb/game with a standard error of 25²⁰. A third party holding only the hand histories cannot reach significance on the win rate, let alone on how fast the agent adapts.

14.5 The joint protocol on Kuhn and Leduc

Seven agents were run against every sub-optimal stationary member of the zoo, against three opponents that switch strategy at hand 1,000 of 2,000, and against three teaching attacks. A teaching attack plays a bait strategy until hand 1,000 and then the exact best response to the agent’s current policy, refreshed every 50 hands — the “taught then exploited” scenario that motivates adaptation safety²². Every condition was run for 10 seeds in both seats (`adaptation_{kuhn,leduc}.json`).

Table 1: Gain against the sub-optimal population, mean exposure of the policy in force, and loss below v^* after a teaching attack, on Kuhn (K) and Leduc (L), chips per hand. 95 % intervals on the gains are at most 0.007.

Agent	Gain K	Exposure K	Teach loss K	Gain L	Exposure L	Teach loss L
Nash	0	0.000	0.000	0	0.000	0.000
BestEq	+0.020	0.000	0.000	+0.047	0.000	0.000
RNR(0.5)	+0.210	0.090	0.045	+0.797	0.264	0.179
DirBR	+0.273	0.338	0.246	+0.988	2.461	1.241
DirBR-CP	+0.230	0.305	0.199	+0.441	2.459	1.757
TypeBR	+0.229	0.352	0.337	+0.961	1.885	2.000

²²Ge, Z., Xu, Z., Ding, T., Meng, L., An, B., Li, W. & Gao, Y. (2024). “Safe and Robust Subgame Exploitation in Imperfect Information Games.” *ICML*, PMLR 235, 15255–15270 — adaptation safety and the “taught and exploited” problem.

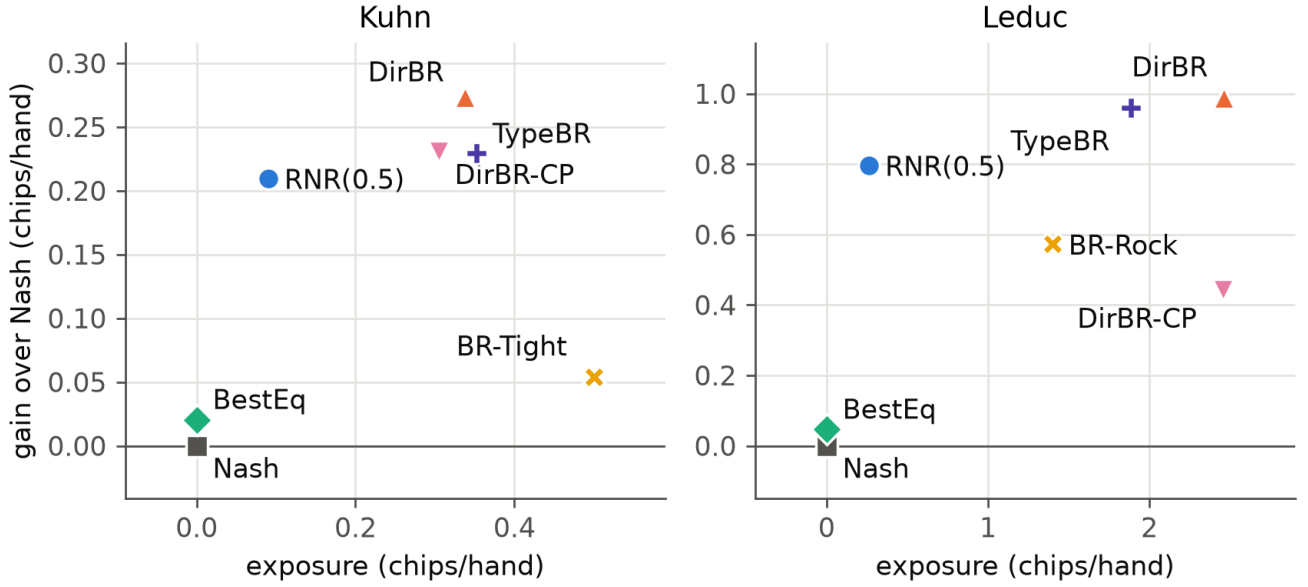


Figure 5: Gain over the Nash blueprint against the sub-optimal population, against exposure. Error bars are 95 % intervals over 10 seeds and are smaller than the markers.

Read together, the numbers separate agents that every single ranking conflates. **BestEq** is safe by construction and gains almost nothing: 0.020 and 0.047 chips/hand. **RNR(0.5)** gains 10 to 17 times as much at an exposure of 0.090 on Kuhn and 0.264 on Leduc, and loses at most 0.274 chips/hand to any teaching attack. **DirBR** gains a little more still (0.988 on Leduc) but at an exposure of 2.46 — almost thirty times the game value — and a teaching attacker takes 1.0–1.5 chips/hand back. Exploitability ranks BestEq, with Nash, first and RNR(0.5) behind them; Elo ranks them the other way. Only the pair (gain, exposure) says what an evaluator needs to know: RNR(0.5) buys most of the gain for a small, measured risk.

Exposure is also the upper envelope of the teaching loss. A white-box attacker in step with the agent’s refits takes exactly the exposure of the current policy. For every agent except DirBR-CP, whose resets leave the attacker’s response stale, the measured teaching losses equal the exposure after the switch.

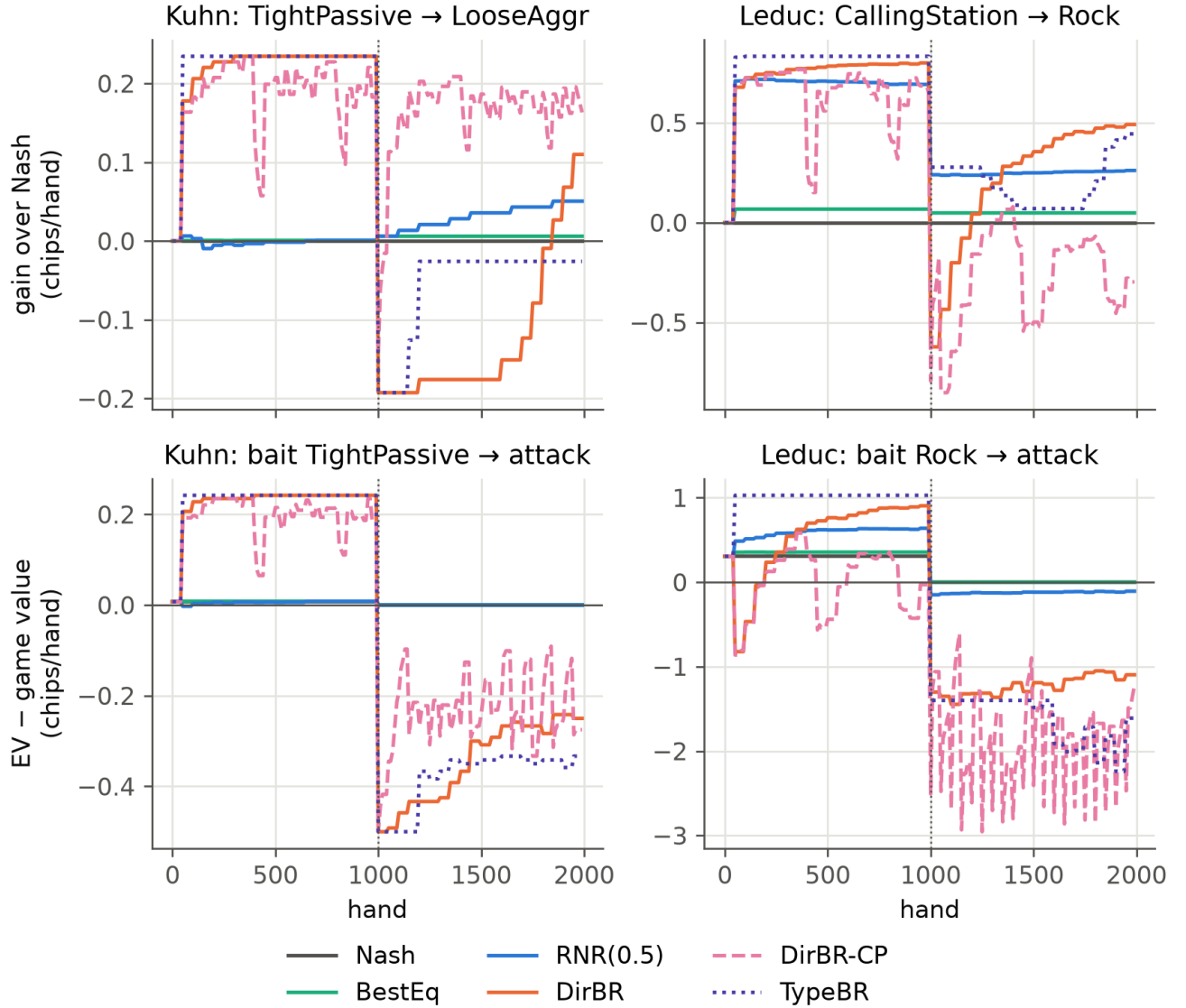


Figure 6: Top: gain over Nash when the opponent switches at hand 1,000 (Kuhn: TightPassive to LooseAggr; Leduc: CallingStation to Rock). Bottom: expected payoff minus the game value under a teaching attack — bait until hand 1,000, then the exact best response to the agent’s current policy, refreshed every 50 hands.

Speed and recovery separate agents with similar gains. On Kuhn every adaptive agent reaches half the attainable gain at the first refit (median $h_{50} = 50$ hands), but in different shares of matches: DirBR-CP 100 %, DirBR 99 %, TypeBR 89 %, RNR(0.5) 73 %. Recovery is informative only where the old exploit fails against the new opponent. On Kuhn the best response to TightPassive does worse than Nash against LooseAggr, so that switch is a real test. There DirBR-CP recovers in a median of 50 hands and RNR(0.5) in 400 (reached in 30 % of matches); DirBR needs 850 (45 %), and TypeBR, whose posterior has absorbed 1,000 hands of TightPassive, never recovers within the match. The change-point exploiter shows why one game is not enough. On Kuhn it recovers fastest and loses least to teaching among the best-response agents. On Leduc its detector fires on *stationary* opponents, it captures only 0.34 of the attainable gain,

and it loses 1.76 chips/hand to the teaching attack against DirBR’s 1.24.

14.6 Approximate best responses and the RRPS score

In games too large to traverse, exploitability is estimated by learning a best response, which gives a lower bound¹¹²³. Kuhn and Leduc allow this to be checked against the exact value.

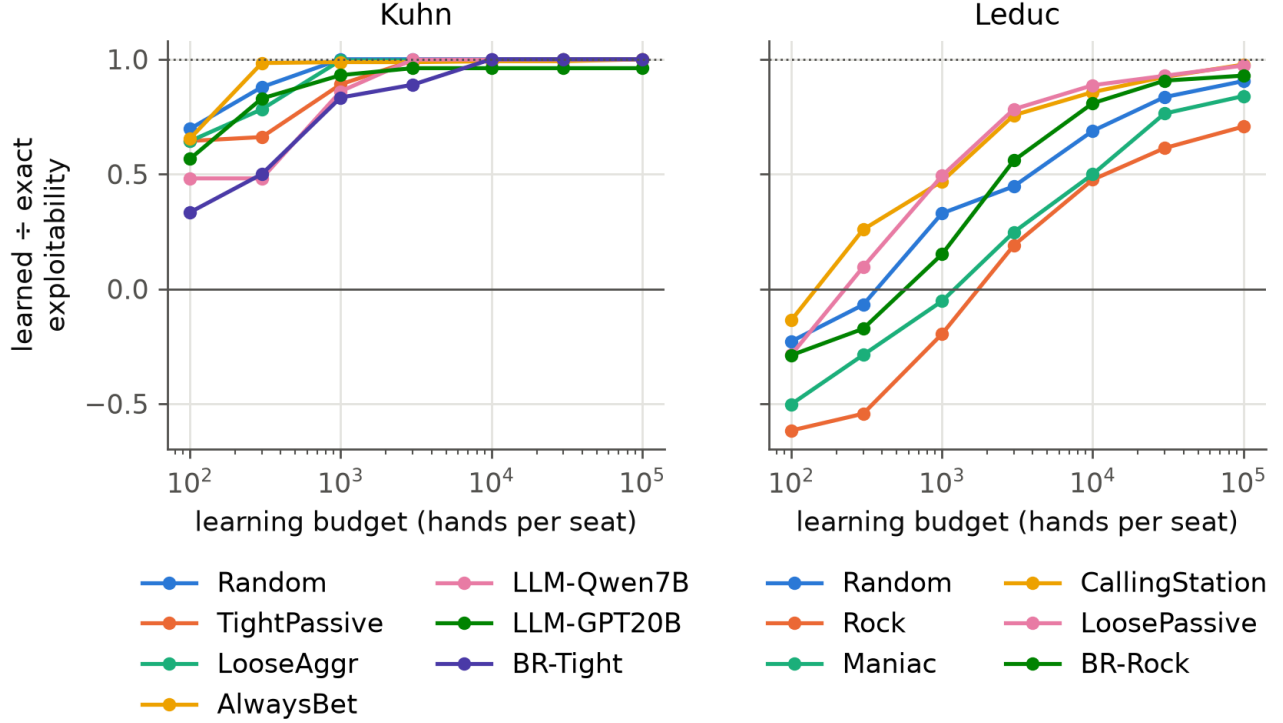


Figure 7: Exploitability found by a learned best response (tabular Monte-Carlo control, $\epsilon = 0.1$) divided by the exact exploitability, as the learning budget grows; mean over 3 seeds and both seats.

On Kuhn the learner comes within 4 % of the exact value by 10⁴ hands for every target. On Leduc it reaches 71 % (Rock) to 98 % (CallingStation) after 10⁵ hands per seat. After 10³ hands it *loses* to Rock and Maniac, so a budget-limited evaluator would report both as safe. Against the adaptive agents the gap is starker. The same learner, playing a 2,000-hand match online, inflicts no damage on Leduc: the agents earn 0.35–0.86 chips/hand above the game value against it, while the white-box attack takes 1.0–1.5 from DirBR. The RRPS benchmark’s *within-population* exploitability — the most any population member wins against the agent — has the same blind spot. It is 0.279 chips/hand for DirBR on Leduc against an exposure of 2.461, and it ranks DirBR second of twelve on the aggregate score. The adversarial-policy results against superhuman Go programs are the large-scale version of this finding²⁴. An exploiter’s

²³Lisý, V. & Bowling, M. (2017). “Equilibrium Approximation Quality of Current No-Limit Poker Bots.” *AAAI-17 Workshop on Computer Poker and Imperfect Information Games*; arXiv:1612.07547 — local best response, a lower bound on exploitability.

²⁴Wang, T. T., Gleave, A., Tseng, T., Pelrine, K., Belrose, N., Miller, J., Dennis, M. D., Duan, Y., Pogrebniaik, V., Levine,

number means little without its budget and strength.

14.7 Three and four players

With three players the protocol changes in one place, the worst case. Five approximate equilibria of three-player Kuhn were computed: CFR and CFR+ (100,000 iterations; NashConv $4.4\text{e-}5$ and $1.5\text{e-}7$) and external-sampling MCCFR with three seeds (NashConv $3.4\text{e-}3$ to $1.4\text{e-}2$). All have near-zero NashConv, yet they are not interchangeable. Seat 0 earns -0.0287 under CFR and -0.0269 under CFR+, and seat 2 correspondingly less, while seat 1 stays at -0.0208 — utility moving between two players inside the equilibrium set, as Szafron et al. describe for this game⁹ — and profiles combining components from different solvers reach a NashConv of 0.125. Worse, an equilibrium component guarantees nothing against a pair. Its coalition value is computed exactly by enumerating one member’s 2^{16} pure strategies and best-responding with the other; a linear objective over correlated joint strategies is minimised at a pure pair. For the CFR and CFR+ equilibria it ranges from -0.094 to -0.125 chips/hand, against equilibrium values of -0.029 to $+0.050$ (nplayer_kuhn3.json).

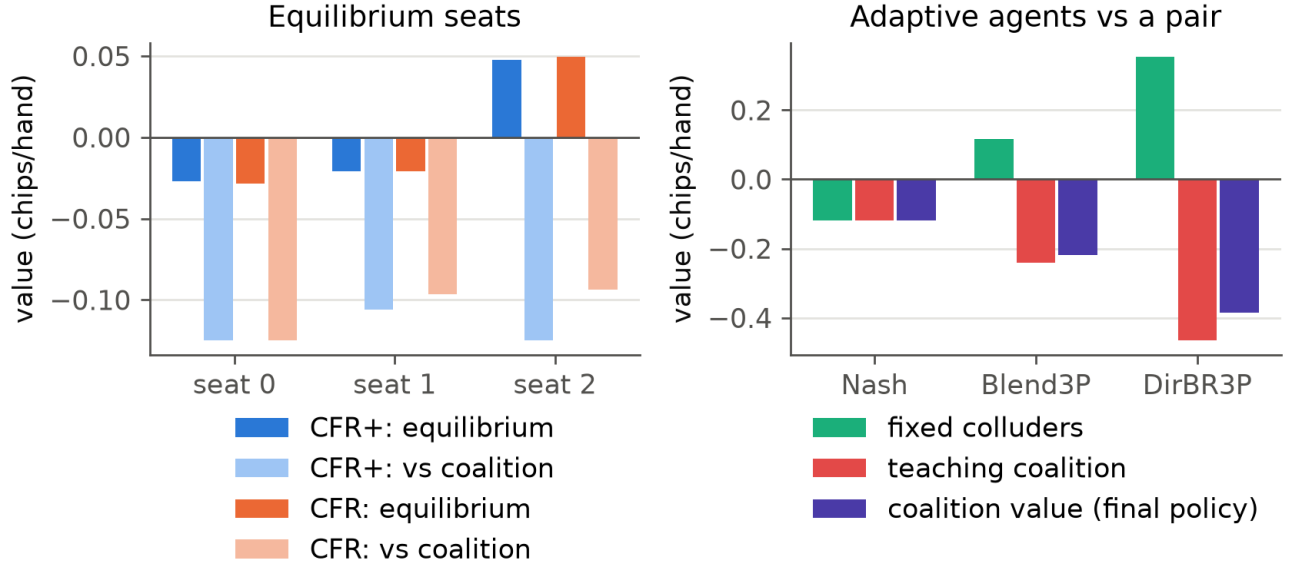


Figure 8: Left: each seat’s value in the CFR+ and CFR equilibria and its coalition value (the worst a coordinated pair can do to that seat’s equilibrium strategy). Right: the mean value of Nash, Blend3P and DirBR3P against a fixed colluding pair (the coalition best response to the blueprint), under a coalition teaching attack after the switch, and the coalition value of each agent’s final policy; 3 seats $\times$ 5 seeds.

The three-player adaptive agent DirBR3P extends Chapter 7’s continuous model to two opponents, spreading counts over the unseen cards by their posterior weight, and best-responds to the modelled pair. Against independent sub-optimal pairs it gains $+0.372 \pm 0.038$ chips/hand over the blueprint and captures 0.99

S. & Russell, S. (2023). “Adversarial Policies Beat Superhuman Go AIs.” *ICML*, PMLR 202, 35655–35739.

of the attainable gain within 50 hands. Most population views put it first: population return (+0.346), a win-share Elo (1972), α -Rank at $\alpha = 0.1$, and Nash averaging of the pairwise projection, which gives it all the mass. The coalition-aware readout disagrees. Against two opponents that bait until hand 1,000 and then play the exact coalition best response to its current policy, DirBR3P earns -0.466 chips/hand, four times the blueprint's -0.119 ; the coalition value of its final policy is -0.386 . Blending with the blueprint (Blend3P) roughly halves both the gain and the teaching loss ($+0.213$; -0.242). A fixed colluding pair aimed at the blueprint costs the blueprint 0.119 per hand but is itself exploitable: DirBR3P earns $+0.354$ against it. Adaptation is a defence against a *static* coalition and a liability against an adaptive one. No ranking built from independently seated opponents can tell these apart, which is failure mode 8 in the gap analysis.

So Long Sucker, re-measured on Chapter 11's engine with its fixed tie-break, supports only the population layer. The pairwise projection is 0.955 transitive, Elo spreads the four baselines over 66 points, and Nash averaging picks the betrayer. A planted alliance lowers a focal baseline's win rate by only 0.8–1.9 percentage points, with intervals that separate for one baseline of four. Chapter 11 found that about 99.5 % of random games in this engine end in deadlock, which leaves an alliance little to act on.

14.8 Where existing evaluation breaks — a checklist

Table 2: The failure modes of `lit_evaluation.md` (point 6, generalisation suites, concerns cooperative substrates and is not tested here) and what this chapter measured.

Failure mode (gap analysis)	Evidence in this chapter	What the protocol reports instead
1. Exploitability only punishes adaptation	BestEq ties Nash for first on exploitability with 0.02–0.05 gain	gain and exposure as a pair
2. No N-player meaning of exploitability	NashConv ≈ 0 components lose 0.06–0.17 more to a pair	the exact coalition value
3. Approximate BRs are lower bounds	learner negative after 10^3 hands; online learner finds nothing	exact exposure; exploiter budget stated
4. Rankings over- or under-credit exploitation	clones, α , horizon change the winner	rankings reported as context
5. Variance compounds with adaptation	3,383 vs 379 hands per window; Pluribus ± 173	estimator named; AIVAT where possible
7. Deceptive opponents untested	teaching attack takes 1.0–2.2 chips/hand on Leduc	teaching loss

Failure mode (gap analysis)	Evidence in this chapter	What the protocol reports instead
8. Collusion evaluated apart from play	DirBR3P first in return, Elo and Nash averaging; -0.47 under a coalition	coalition teaching attack
9. LLM arenas inherit the above	Elo ranks the Qwen strategy above a less exploitable bot	the same readouts for LLM agents

14.9 Honest notes, limitations, and where this hands off

What held up. Every exact quantity matches an independent reference: OpenSpiel’s NashConv, α -Rank, Nash averaging and maximal lotteries; Chapter 3’s exploitability; Chapter 7’s best responses; Chapter 8’s solvers. AIVAT is exactly unbiased, and its zero-variance limit now holds. The protocol’s gains carry 95 % intervals of at most 0.007 chips/hand on the two-player games, because the policy-exact estimator has no card noise; the ten seeds sample the only remaining randomness, the learning trajectory.

What did not, and why it matters. Three plan predictions failed and are kept as stated: “random loses to everyone” (on Kuhn it beats AlwaysPass); “AIVAT $\geq 10\times$ on Leduc” (7 of 12 pairs); “approximate within 10 % on Leduc” (4 of 6 targets, only at 10^5 hands). Two of my own implementation errors were caught by checks, not by luck: the AIVAT double count, and a Nash averaging solver that returned infeasible “optimal” points on near-tied agents until a fixed tolerance and support LPs replaced it.

Limitations. These are toy games. The exact worst case and the exact coalition value (2^{16} strategies per member) exist only because the games are tiny; at scale a learned attacker gives only a lower bound on either. The policy-exact estimator exists only for bots. The white-box teaching attacker is a strong adversary but not the worst adaptive one. The adaptive agents are Chapter 7–8 baselines: the protocol *measures* the missing N-player loss bound; it does not provide one. So Long Sucker is weak evidence. The teaching attacker refreshes every 50 hands, in step with these agents’ refits; against an agent whose policy changes every hand, that understates the attack.

Connections. Backward: the engine absorbs Chapter 3’s exploitability, Chapter 7’s models and best responses, Chapter 8’s safe responses (now as one LP), Chapter 10’s spinning top, Chapter 11’s So Long Sucker and Chapter 12’s LLM-extracted strategies. Forward: for Contribution 2 the protocol supplies the measurement a safe N-player agent must pass — gain against independent pairs, coalition value, and coalition teaching loss. The one-shot LP removes the scaling obstacle Chapter 8 met.

14.10 Key takeaways for the thesis synthesis

- **Exploitability and rankings answer different questions, and for exploiting agents they give opposite answers.** On the same Leduc zoo, BestEq and Nash lead exploitability and Nash averaging and BestEq leads VasE; RNR(0.5) and DirBR lead Elo, population return and the RRPS score. The Elo–exploitability rank correlation is 0.36.
- **Gain, speed, recovery and exposure reported together separate what single numbers conflate.** RNR(0.5) gains +0.797 chips/hand on Leduc at exposure 0.264 and at most 0.274 teaching loss; DirBR gains +0.988 at exposure 2.461 and 1.0–1.5 teaching loss; BestEq gains 0.047 at zero.
- **With three players the worst case must be coalition-aware.** Equilibrium components (Nash-Conv ≈ 0) lose 0.06–0.17 chips/hand more to a coordinated pair than they earn in equilibrium. The adaptive DirBR3P tops population return, Elo and Nash averaging, and loses 0.47 chips/hand to a coalition teaching attack.
- **Confidence is the binding constraint on measuring adaptation.** Per-window estimates need about 3,400 hands with raw chips against about 380 with AIVAT on Kuhn. On Pluribus’s released hands the raw 95 % interval is ± 173 mbb/hand.
- **Reported exploiters need a stated budget.** A learned best response reports two exploitable Leduc bots as safe after 10^3 hands, and within-population exploitability misses DirBR’s exposure by a factor of nine.
- **The C3 claim stays narrow.** The pieces exist; what this chapter adds is their joint, unchanged application across two- and three-player imperfect-information games, and a demonstration of where each existing measure breaks. All of it is on toy games so far.